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Syllabus: Description,
Learning Objectives, and
Grading

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10 min

Hypothesis Testing

Power

Two Sample Tests

Review

Syllabus

Course Description

This class presents fundamental concepts in data analysis and statistical inference, focusing on one and two independent samples. Students having taken this class should be able to summarize samples, perform relevant hypothesis tests and perform a collection of two sample comparisons. Classical non-parametric methods and discrete data analysis methods are discussed. The class is taught at a master's of biostatistics introductory level and requires Mathematical Biostatistics Boot Camp 1 as a prerequisite.

Learning Objectives

The goal of this course is to equip biostatistics and quantitative scientists with core applied statistical concepts and methods:

- The course will refresh the mathematical, computational, statistical and probability background that students will need to take the course.
- The course will introduce students to the display and communication of statistical data. This will include graphical and exploratory data analysis using tools like scatterplots, boxplots and the display of multivariate data. In this objective, students will be required to write extensively.
- Students will learn the distinctions between the fundamental paradigms underlying statistical methodology.
- Students will learn the basics of maximum likelihood.
- Students will learn the basics of frequentist methods: hypothesis testing, confidence intervals.
- Students will learn basic Bayesian techniques, interpretation and prior specification.
- Students will learn the creation and interpretation of P values.
- Students will learn estimation, testing and interpretation for single group summaries such as means, medians, variances, correlations and rates.
- Students will learn estimation, testing and interpretation for two group comparisons such as odds ratios, relative risks and risk differences.
- Students will learn the basic concepts of ANOVA.

Grading Policy

- Grades are based on student performance on the 4 Module Quizzes. To pass the course, you must earn an 80% or better on each of the four graded quizzes. Students who pass the course and verify their identity will earn a course certificate.
- For calculation of your final grade, the quizzes are weighted as follows: Module 1 Quiz, 38%; Module 2 Quiz, 18%; Module 3 Quiz, 23%; Module 4 Quiz, 20%
- Module homework assignments are for practice only. Although they are graded, your homework grades will have no impact your final course grade. You are encouraged to complete the homework assignments because they will help you master the material covered within this course.

Discussion Forums

All students are encouraged to participate in the Discussion Forums by asking and answering questions and providing feedback.

The instructor will also participate in the Forums by answering the questions that receive a large number of up-votes or are generating a considerable amount of discussion. The instructor is not able to personally respond to every questions, so students are strongly encouraged to help one another by answering questions and by up-voting the questions that they would like to see Dr. Caffo answer.

All students are expected to be civil when participating in the Discussion Forums.

- Be friendly and considerate when talking to your fellow students.
- Use up-votes to bring attention to thoughtful, helpful posts.
- Search before you post.
- Post in the appropriate sub-forum.
- Use the flag icon to report inappropriate content or highlight posts related to platform issues.

Go to next item ✓ Completed

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